

WebCrawler

Functional requirements:

- WebCrawler should be able to download from particular file,
- WebCrawler should be able to place multiple downloads ,
- WebCrawler should be able to handle network failure scenario as well

Non-functional requirements:

- Data-partitioning,
- load balancing,
- fault tolerance,
- vertical scalability,
- horizontal scalability,
- low latency,
- high throughput,
- CAP theorem (Here we eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:

- user ,
- file,
- history,

API :

POST:/download
{
 userID,
 filemetadata
}
GET:/search/{fileID}

HLD :

